

Input
$\delta\mathbf{F}_c$, u
(controls, iteration i)

(1) Differentiable MLS-MPM

Forward simulation ($2T$ steps)
- driven half: $\delta\mathbf{F}_c$ and u act
- released half: controls set to zero
- P2G / grid / G2P
- outer-layer relaxation

Physics loss on the released end state
- Sinkhorn transport to the target
- proximity, end drift, released motion

(2) Exterior Surface

Surface from the particles
- Zhu–Bridson field of the particles
- exterior discs on its zero level
- normals from the field

Differentiable
- the discs move with the particles, so render gradients reach them

(3) Differentiable Rendering

Render
- soft splats of the discs
- 18 orthographic views
- silhouette and Lambert shading

Render loss
- $\mathcal{L}_{\text{sil}} + \mathcal{L}_{\text{shade}}$ against the pictures of the target (the same 18 views)

$\nabla\mathcal{L}_{\text{phys}}$

$\nabla\mathcal{L}_{\text{render}}$

Adjoint through the $2T$ steps (reverse mode of the same MLS-MPM)

$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{phys}} + \lambda\,\mathcal{L}_{\text{render}}$

PCGrad drops the part of $\nabla\mathcal{L}_{\text{render}}$ that opposes $\nabla\mathcal{L}_{\text{phys}}$; $\quad \lambda\,\|\nabla\mathcal{L}_{\text{render}}\| = \frac{1}{2}\|\nabla\mathcal{L}_{\text{phys}}\|$

inject $\nabla\mathcal{L}_{\text{total}}$

Control update
Adam step with line search;
$\delta\mathbf{F}_c$ capped;
u free of rigid modes